SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.3: Rotation — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.3: Rotation
Driving Question	DRIVING QUESTION: When the minute hand of the clock at the University of Nairobi completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up — and how can we use those rules to rotate any shape to any position, on paper and on a grid?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Spinning Hands of Nairobi's Clock Tower: What Is Really Happening?**	Using card-arrow clocks (improvised hands pinned at a centre point), learners physically rotated the arrow and tracked chosen points on the hand. They measured the distance from each tracked point to the pin (centre) before and after rotation, and observed that this distance never changed regardless of how far the hand was turned. They also noted that the hand swept through a measurable angle and that the direction of turning (clockwise or anticlockwise) determined where the tip ended up.	Every point on a rotating object remains exactly the same distance from the centre of rotation before and after the turn. A rotation is fully described by three things: (1) the centre of rotation — the fixed pivot point, (2) the angle of rotation — how many degrees the object turns, and (3) the direction of rotation — clockwise (CW) or anticlockwise (ACW). These three elements are the minimum information needed to define any rotation uniquely.	Pedagogical note: This lesson anchors the entire sub-strand to the driving question. When learners measure equal radii before and after turning, they are building the intuition that rotation is a distance-preserving transformation — a fact they will formalise in Lesson 2. Ensure every learner records the three descriptors (centre, angle, direction) in their notes using the University of Nairobi clock as the concrete referent. Cold-call at least four learners to state all three descriptors aloud before closing. Common misconception to address: some learners will claim the centre itself 'moves slightly' — have them re-measure carefully to correct this.
Lesson 2 **Discovering the Properties of Rotation**	Learners rotated cut-out triangles and quadrilaterals on dot paper about a marked centre, then traced both the object and its image. They measured corresponding sides and angles of the object and image using rulers and protractors, and compared the distances of corresponding vertices from the centre of rotation. They also examined	Rotation has four properties that learners co-constructed: (1) Congruence — the image is exactly the same shape and size as the object; all corresponding sides and angles are equal. (2) Equidistance from centre — every object point and its corresponding image point are the same distance from the centre of rotation. (3)	Pedagogical note: Property 3 is the key bridge to the coordinate rules learners will use in Lessons 3–4. Emphasise that the angle at the centre between OA and OA′ is always equal to the stated angle of rotation — this is what makes rotation predictable and rule-governed. Use the clock analogy: the minute hand sweeps the same angle

	whether the cyclic order of vertices (going around the shape) changed after rotation.	Equal angles at centre — the angle formed at the centre between any object point and its image point equals the angle of rotation. (4) Opposite (indirect) congruence — the cyclic order of vertices is reversed in the image, so the image cannot be superimposed on the object by sliding alone; a physical flip would be needed if the rotation were 'unwound'.	from every marked minute position to the next. Provide wait time of at least 30 seconds after asking 'Why must the image be congruent to the object?' before cold-calling. Target learners who measured results but have not yet articulated the reason (distance preservation).
Lesson 3 **Rotating Shapes on Paper: Constructing Rotation with Compass and Protractor**	Learners used a compass and protractor to rotate given triangles and quadrilaterals about a specified centre (both inside and outside the shape) through given angles (90°, 120°, 180°) in specified directions. For each vertex, they (i) drew a ray from the centre through the vertex, (ii) used the protractor to measure the required angle in the correct direction, (iii) used the compass to mark the image vertex at the same radius, then connected image vertices to complete the image shape.	The formal paper-construction procedure for rotating a point P about centre O through angle θ is: (1) draw segment OP; (2) at O, use a protractor to construct an angle of θ in the required direction from OP; (3) mark P′ on the new ray at distance OP (using a compass set to radius OP). Repeat for every vertex, then join the image vertices in order. The image produced is congruent to the object and every image vertex lies on the same circle (centred at O) as its corresponding object vertex. This procedure works for any centre, any angle, and any direction.	Pedagogical note: Many learners confuse the direction of angle measurement — stress that CW means the image ray is to the right of the object ray when viewed from above, matching the direction the University of Nairobi clock hands move. A common error is resetting the compass between vertices; remind learners to keep the compass opening fixed only when copying the same radius. Circulate and check that learners measure the angle at O (not at the vertex). After practice, ask: 'How does this construction connect back to Property 3 from Lesson 2?' — this reinforces coherence across lessons.
Lesson 4 **Rotation on the Cartesian Plane: Coordinate Rules**	Learners plotted simple shapes (triangles with integer coordinates) on squared-paper Cartesian grids, then applied rotation rules to generate image coordinates without a protractor. They verified each image by also constructing it with a protractor/compass, confirming both methods give the same result. The four rules investigated were: 90° ACW (or 270° CW): (x, y) → (−y, x); 90° CW (or 270° ACW): (x, y) → (y, −x); 180° (either direction): (x, y) → (−x, −y); 360°: (x, y) → (x, y).	For rotations about the origin O(0, 0), four coordinate transformation rules apply: (1) 90° anticlockwise: (x, y) → (−y, x). (2) 90° clockwise: (x, y) → (y, −x). (3) 180°: (x, y) → (−x, −y). (4) 360°: (x, y) → (x, y) — the identity, returning every point to itself. These rules are exact and eliminate the measurement error inherent in protractor constructions. When the centre is not the origin, learners translate the centre to the origin, apply the rule, then translate back. The rules are consistent with the properties established in Lesson 2: distances from the origin are preserved because $(-y)^2 + x^2 = x^2 + y^2$.	Pedagogical note: Connect 360° explicitly to the driving question — when the University of Nairobi minute hand completes a full journey, every point returns to (x, y): mathematically, nothing has changed in position. Use this to discuss why 360° = 0° as a rotation. A powerful diagnostic question: 'A point is at (3, 4). After a 90° CW rotation about the origin, where is it? Check your answer using the distance formula.' (Answer: (4, −3); distance from origin = 5 both times.) Cold-call at least five learners with different starting points to build fluency before setting independent practice.
Lesson 5 **INVESTIGATE: Finding the Centre and Angle of Rotation**	Given only an object and its image (no centre marked), learners used the perpendicular bisector method to locate the centre of rotation: they drew perpendicular bisectors of segments joining at least two pairs of corresponding vertices and	The centre of rotation is the unique point equidistant from each object vertex and its corresponding image vertex. It therefore lies on the perpendicular bisector of the segment joining every object–image vertex pair. In practice: (i) draw segments AA′ and	Pedagogical note: This lesson develops the NGSS practice of 'analysing and interpreting data' — learners work backwards from evidence (the object and image) to reconstruct the transformation. Emphasise accuracy in bisector

	identified their intersection as the centre O. They then measured the angle at O between one object–image vertex pair to determine the angle of rotation, and used the vertex order to determine direction.	BB′ (joining corresponding vertices); (ii) construct the perpendicular bisector of each segment; (iii) their intersection is the centre O. The angle of rotation equals ∠AOA′ (measured at O), and the direction is determined by whether the image vertices are arranged clockwise or anticlockwise relative to the object vertices when viewed from O. This is the reverse problem of Lessons 3–4 and completes the bidirectional understanding of rotation.	construction; a 1 mm error in drawing the perpendicular bisector can shift the apparent centre significantly. A useful check: measure OA, OA′, OB, OB′ — all four should be equal. Connect to the driving question: 'If you saw a photograph of the University of Nairobi clock at two different times but the face was blank, how would you find the angle the hand had rotated?' This makes the investigation feel purposeful.
Lesson 6 **Rotational Symmetry of Plane Figures and Solids**	Learners rotated cut-outs and 3-D models (cubes, square-based pyramids, cylinders) about their centres of symmetry and counted how many times the shape mapped exactly onto itself within a full 360° turn. They recorded results in a table and derived the minimum angle of rotational symmetry for each figure. Kenyan cultural objects — including the shape of a Maasai shield (approximate bilateral/rotational symmetry), a circular ugali plate, and a hexagonal beehive cell — were included as reference shapes.	A plane figure has rotational symmetry of order n if it maps onto itself exactly n times during a full 360° rotation about its centre. The minimum angle of rotation for each mapping is 360°/n. Key results: equilateral triangle — order 3, minimum angle 120°; square — order 4, minimum angle 90°; regular hexagon — order 6, minimum angle 60°; circle — infinite order; a scalene triangle — order 1 (no rotational symmetry beyond 360°). For 3-D solids, rotational symmetry axes are lines about which the solid maps onto itself; a cube has 13 axes of rotational symmetry. Order 1 means no non-trivial rotational symmetry.	Pedagogical note: The formula minimum angle = 360°/n is simple but frequently misapplied — learners sometimes use n = number of sides rather than verifying by counting actual self-mappings. Insist on the counting-first approach before applying the formula. A strong cross-curricular link: Kenya's flag has a Maasai shield at its centre — ask learners to determine its order of rotational symmetry (answer: order 1 for the shield shape itself, but the circular symmetry of the surrounding design is order 1 due to colour asymmetry). This grounds the abstract concept in national identity and cultural relevance.
Lesson 7 **Rotation and Congruence: Same Shape, Different Story**	Learners rotated a scalene triangle through various angles and compared the object and image by attempting to slide (translate) the image onto the object without lifting or flipping it. They found that no slide alone could superimpose image on object for rotations other than 0°/360°. They then compared rotation with reflection outputs, noting that both produce congruent figures but that reflection produces a mirror-image (laterally reversed) figure while rotation does not — yet both are classified as 'opposite congruence' in formal geometry because cyclic vertex order is reversed.	Rotation produces opposite (indirect) congruence: the image has the same side lengths and angle measures as the object (it is congruent), but the cyclic order of corresponding vertices is reversed. This means a rotated image cannot be mapped onto the original object by a translation alone — a rotation back or a combination of transformations is required. Crucially, rotation does NOT produce a mirror image: if you write 'Nairobi' and rotate it 180°, the letters appear upside-down but not laterally reversed (unlike a reflection). This distinguishes rotation from reflection while both remain congruence-preserving transformations.	Pedagogical note: The 'Nairobi' written-word demonstration is highly effective — write the word on a card, rotate the card 180°, and compare with its reflection in a mirror. Learners viscerally see the difference. Emphasise that in the Kenyan national examinations, learners are often asked to classify transformations given an object and image: opposite congruence narrows it to rotation or reflection, and the absence of a mirror line (or the presence of a clear centre) distinguishes them. Provide at least three exam-style classification questions as formative exit tasks, cold-calling learners to justify their classification with reference to vertex order.
Lesson 8 **Model Building, Application &**	Learners revisited their cumulative rotation model (built across Lessons 1–7) and completed a structured application task set	All seven Sub-Strand 2.3 SLOs were consolidated: (a) Rotation on the Cartesian plane — 90° CW: (x,y)→(y,−x); 90° ACW:	Pedagogical note: This lesson serves simultaneously as a synthesis, model-revision, and formative assessment

Assessment: "Putting It All Together"	in the context of the University of Nairobi clock: given the positions of points on the minute hand at different times, they (i) applied coordinate rules to find image positions, (ii) used the perpendicular bisector method to find a centre of rotation, (iii) identified rotational symmetry orders of clock-face numerals and the clock's overall design, and (iv) classified the transformation as rotation vs. reflection using congruence reasoning.	$(x,y) \to (-y,x)$; 180°: $(x,y) \to (-x,-y)$; 360°: $(x,y) \to (x,y)$. (b) Rotating shapes on paper using compass and protractor. (c) The three descriptors that define a rotation: centre, angle, direction. (d) The four properties of rotation: congruence, equidistance from centre, equal angles at centre, opposite congruence. (e) Finding the centre and angle of rotation given object and image. (f) Rotational symmetry — order n, minimum angle = 360°/n — for plane figures and solids. (g) Rotation produces opposite congruence, distinguishable from reflection by the absence of a mirror line and the direction of vertex-order reversal.	session. Use the completed model as a talking-piece: ask each group to identify which part of their model changed most from Lesson 1 to Lesson 8 and why. The driving question should now be fully answerable — cold-call four learners to give a complete verbal answer referencing centre, angle, direction, coordinate rules, and congruence. For differentiation, extension learners can explore what happens when two successive 90° CW rotations about different centres are combined (result: a single rotation or translation), previewing Lesson content in later sub-strands. Collect written answers to the driving question as a summative artefact.